

Robot Framework: Test Automation Made Quick and Easy

The feature-rich Robot Framework can assist us in the design and development of mature end-to-end automation solutions, with ease. As the scripts are keyword based, they are readable, which makes it easy for different stakeholders to collaborate in quality engineering of the product.



The characteristics and components of a good automation framework are:

- An appropriate language like Python, Java, etc, to write the tests and create scripts.
- Reusable pages or entity objects.
- Supporting libraries that offer the capability to test Web, API, database or mobile applications.
- An abstraction layer that provides users the ability to write test cases in a natural language so that it can also be understood by business analysts.
- A test runner that has selective capabilities like tagging test cases and running them by tags.
- Test suites or test setups and teardown mechanisms that enable us to clean up and also not repeat setup and teardown scripts.
- A key component is well structured reports that have a results summary with aggregation, timestamps, the ability to drill down into the test case to see what was executed, as well as the ability to search results and automatically embedded screenshots if there is a failure.
- The capability to schedule tests daily, weekly or monthly.
- The capability to integrate with a continuous integration system so that the tests can be run automatically after the build and deployment on a server.

- The support for parallel execution so that tests can be run on different computers simultaneously to save time.
- Finally, the need to have some sense of convention to bring about consistency within the team.

Why the Robot Framework?

The simple answer to this question is that the Robot Framework does all of these things, right out of the box, so that we do not require to build test automation frameworks from scratch. What's even better is that it's open

source and free. It can also be installed, be up and running, and ready to use in no time and with minimal effort. It's definitely the fastest and easiest test automation framework to learn.

An overview of the Robot Framework libraries

Robot Framework scripts are mostly keyword driven, i.e., we can type in some simple, plain English words to invoke pre-written logic in some pre-existing libraries that have been made available to us. This gives us the ability to create our scripts in a natural

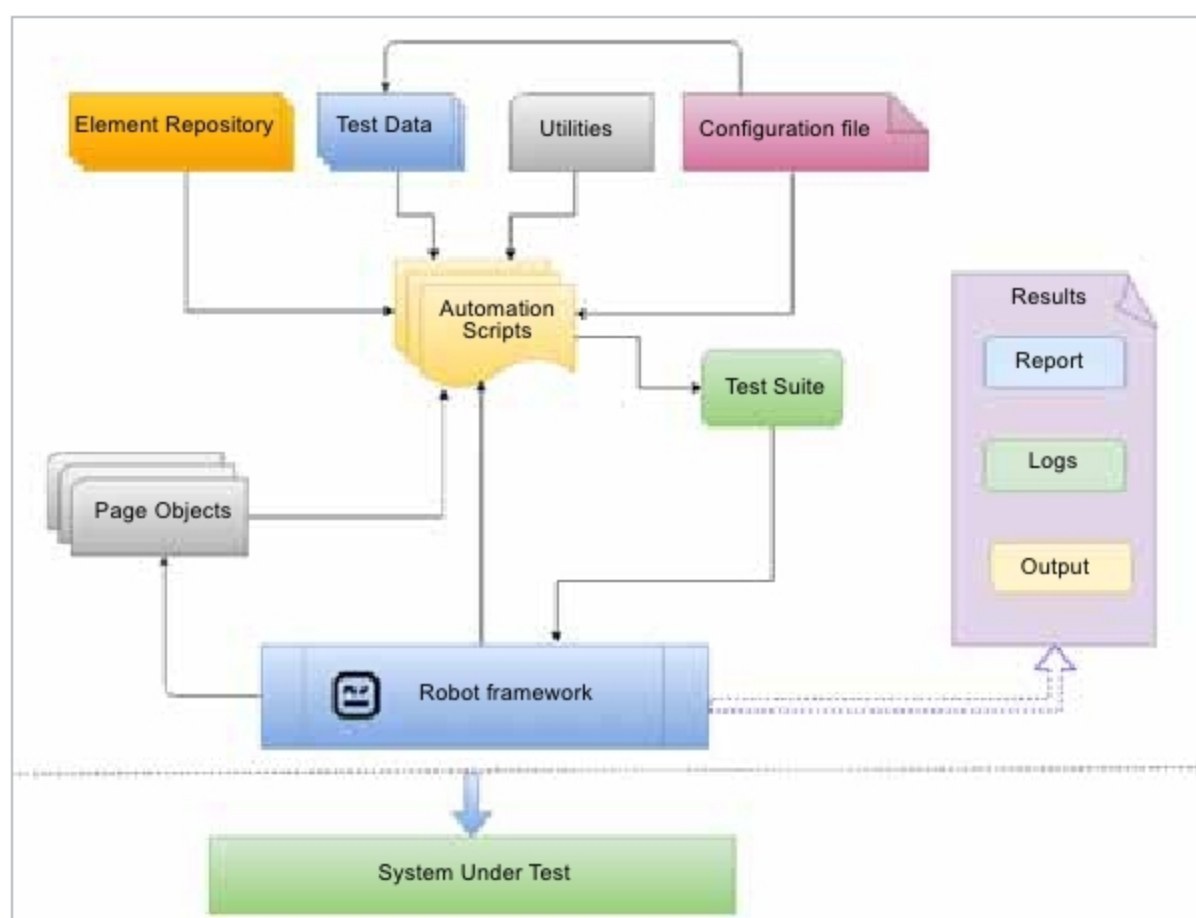


Figure 1: Test automation framework architecture

